

ELE 404: Electronics I

MOSFET Amplifier Design Project

Introduction

Design an open-loop analog voltage amplifier **using MOSFET circuits only**. The design must be verified **entirely through SPICE simulations**. You are free to choose the circuit topology and number of stages, provided all specifications and design rules below are met. This project emphasizes **architectural reasoning, biasing, and device-level trade-offs**. *Note that this project is intentionally open-ended. Multiple architectures can meet the specifications.*

Table I. Design Specifications

Category	Parameter	Specification / Constraint
Technology	Supply voltage	Single supply, VDD = 3.3 V
	MOSFET models	NMOS/PMOS
	Minimum channel length	$L \geq 0.5 \mu\text{m}$
Input	Input type	Voltage input
	Signal amplitude	$\leq 10 \text{ mVpp}$
	Source resistance	50Ω
	Input coupling	AC
	Minimum input resistance	$\geq 100 \text{ k}\Omega$
	Biasing	Resistor divider or alternative biasing techniques
Gain	Total open-loop gain (unloaded)	$\geq 60 \text{ dB}$ ($\geq 1000 \text{ V/V}$)
	Gain per stage	$\leq 40 \text{ dB}$ (justify if exceeded)
Bandwidth	–3 dB bandwidth	$\geq 500 \text{ kHz}$
Output	Output type	Single-ended
	Load resistance	$R_L = 10 \text{ k}\Omega$
	Load capacitance	$C_L = 2 \text{ pF}$
	Output swing	$\geq 1.5 \text{ Vpp}$
	Load sensitivity ($R_L = 10 \text{ k}\Omega$)	Gain reduction $\leq 10\%$
Power	Total DC power	$\leq 1 \text{ mW}$
	Max branch current	$\leq 200 \mu\text{A}$

Table II. Technology Specifications (Global Foundries 180nm CMOS)

Parameter	Description	Unit	Value
V_{DD}	Supply	V	3.3
t_{ox}	Oxide thickness	nm	8
$C_{oxn}=C_{oxp}$	Oxide capacitance	(fF/ μm^2)	4.3
C_{ov}	Overlap capacitance	fF/ μm	0.4
V_{tn}	NMOS threshold voltage	V	0.63
V_{tp}	PMOS threshold voltage	V	-0.73
$\mu_n \cdot C_{ox}$	NMOS process transconductance	A/V ²	100
$\mu_p \cdot C_{ox}$	PMOS process transconductance	A/V ²	25
L_{min}	Minimum length	μm	0.28
λ_n, λ_p	Channel length modulation factor	1/V	≈ 0.04

Design guidelines

- Estimate parameters with hand calculations (expect mismatches of 10-20% with respect to simulated values)
- Use transistor lengths $0.5\mu\text{m} \leq L \leq 5\mu\text{m}$
- Set the overdrives $0.2V \leq V_{ov} \leq 0.6V$
- Run DC sweeps to determine the optimum operating points

Report requirements

Including the cover page, the report is limited to 10 pages.

- Identify and justify the architecture of each amplification stages, which when cascaded will meet the given design requirements (Common-source, common-gate, common-drain).
- Present manual calculations, and explain in sufficient details, the sizing and biasing of all transistors in the circuit.
- Simulate the designed amplifier in KiCad (or any other equivalent circuit simulation software). Demonstrate that the design meets the requirements and that its simulated performance is in a reasonable agreement with those predicted through manual calculations. Report screenshots of simulated waveforms and table with performance results based on given specifications.

Evaluation (detailed marking rubric in D2L)

- Design approach and justification in alignment with theory (30%)
- Simulations of the design showing performance based on parameters in Table I. (30%)
- Discussion of the design (30%): Interpretation of the results, mismatches between simulated performance and results, and discrepancies between simulations and calculations.
- Organization, formatting and clarity of the report (10%)

Table III. Useful formulae

Total transistor gate capacitance (sat), C_{GS}	F	$\approx 2/3 \cdot C_{OX} \cdot W \cdot L$
Total transistor drain capacitance (sat), C_{GD}		$\approx C_{OV} \cdot W$
Single-stage amplifier bandwidth (f_p)	Hz	$\frac{1}{2\pi R_{node} C_{node}}$
Transconductance, g_m	A/V	$\frac{2I_D}{V_{OV}}$